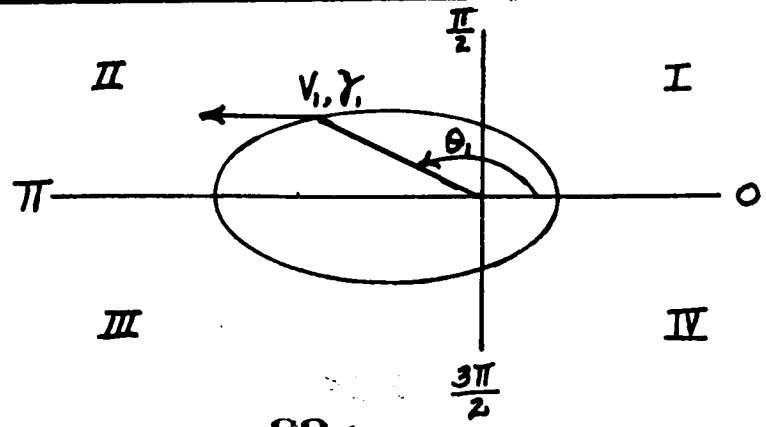
perigee. The vehicle is assumed to be launched as a free body at $\theta = \theta_1$. The initial elevation angle γ_1 is positive when the initial velocity vector V_1 is inclined outward from the perpendicular to the initial radius r_1.

Boundaries separating various values of the orientation angle θ_1 are shown in figure 1-4. The curve shown represents the values of (V_1/V_s) and γ_1 where θ_1 equals 90° or 270° as defined by

$$\frac{V_1}{V_s} \cos \gamma_1 = 1 \qquad (1.5\text{-}23)$$

This curve and the axis where $\gamma_1 = 0$ divide the area into the four quadrants as shown in figure 1-4 and as indicated below:

Conditions	Quadrant
$\gamma +, \quad 1.0 < \frac{V_1}{V_s} \cos \gamma_1 < \sqrt{2} \cos \gamma_1$	I
$\gamma +, \quad 0 < \frac{V_1}{V_s} \cos \gamma_1 < 1.0$	II
$\gamma -, \quad 0 < \frac{V_1}{V_s} \cos \gamma_1 < 1.0$	III
$\gamma -, \quad 1.0 < \frac{V_1}{V_s} \cos \gamma_1 < \sqrt{2} \cos \gamma$	IV



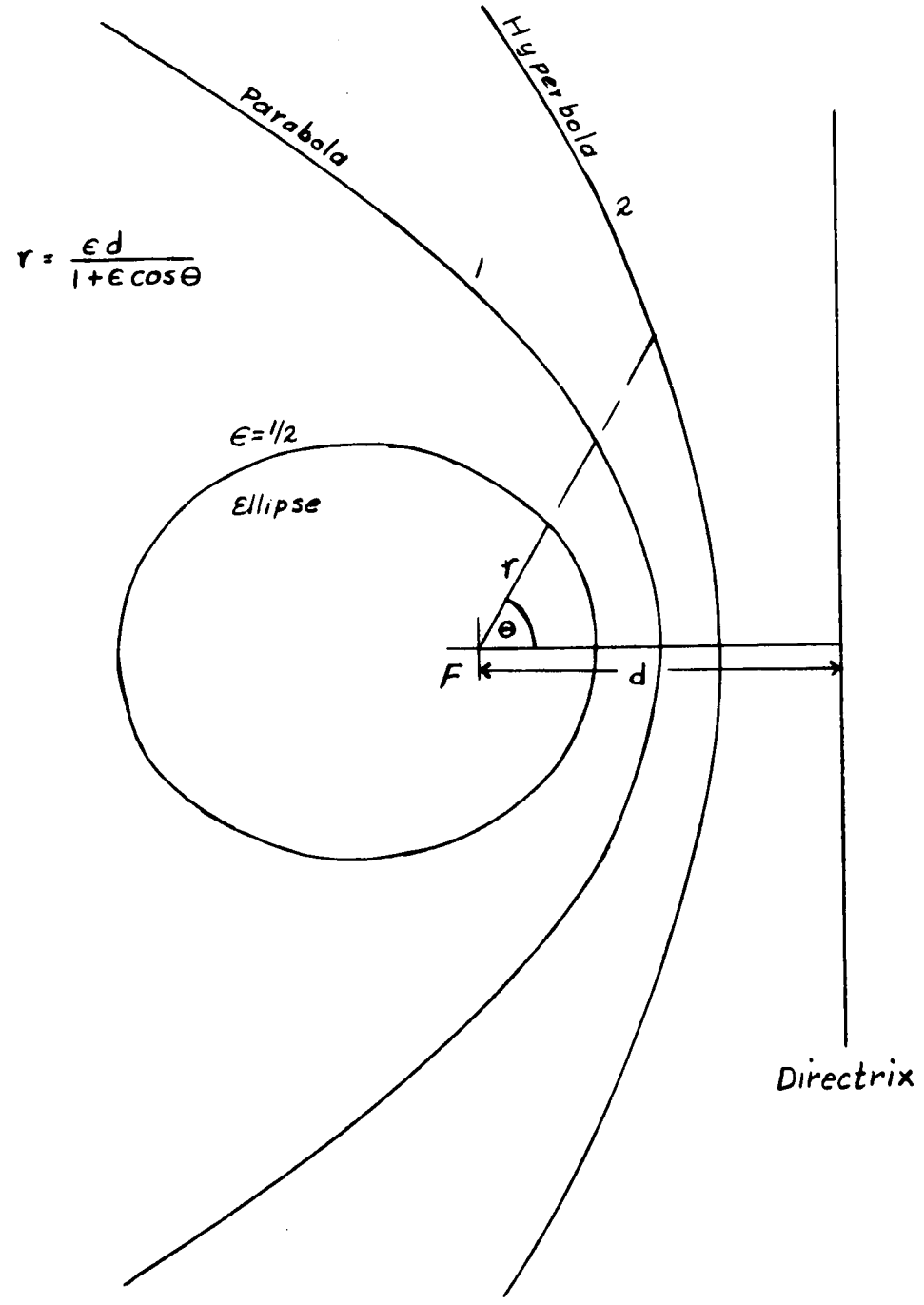


Fig 1-1.- Conic sections, ellipse, parabola, hyperbola

List of Symbols

X_N, Y_N, Z_N	right-hand Cartesian coordinate system fixed in non-rotating earth, inertial axes (with Z_N positive down)
ψ, θ, φ	angles used in specifying missile attitude referred to as Euler angles (specified order of rotation ψ, θ, φ).
x_b, y_b, z_b	body axes - right-handed coordinate system fixed in missile body with the origin at the instantaneous center of gravity. (See Figure 2.)
p, q, r	angular velocities of body-axis system (x_b, y_b, z_b) positive clockwise when looking in positive direction of axes
u, v, w	components of resultant velocity V along body axes x_b, y_b, z_b, respectively
i_b, j_b, k_b	unit vectors along the x_b, y_b, and z_b axes respectively
i_N, j_N, k_N	unit vectors along the X_N, Y_N, and Z_N axes respectively
V	resultant velocity of missile center of gravity
ω	resultant angular velocity of missile or (x_b, y_b, z_b) system $\bar{\omega} = i_b p + j_b q + k_b r$
V_e	jet exit velocity relative to nozzle exit
U	Momentum of missile $\bar{U} = m\bar{V}$
U_T	momentum of missile and jet
m	missile mass (slugs)
H	angular momentum of missile with respect to the center of gravity or moment of momentum
H_T	angular momentum of missile and jet